TENSOR RELATORS ALONG GENERATING SETS OF TWO MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Generating sets give a surjection from a direct sum of copies of the ring. Used at the short exact sequence in the counted proof (lines 26263-26268). Presents N as a quotient of the direct sum over the generating set.
- Right exactness of the tensor product. Used at the tensored sequence in the counted proof (lines 26268-26271). Transports the presentation of N to a presentation of N tensor M, which is where the relation coefficients are read off.

Lemma 0.1. *Let R be a ring. Let M, N be R-modules. Let $\{x_i\}_{i\in I}$ be a set of generators of M. Let $\{y_j\}_{j\in J}$ be a set of generators of N. Let $\{m_j\}_{j\in J}$ be a family of elements of M with $m_j = 0$ for all but finitely many j. Then*

$$\sum\nolimits_{j\in J} m_j \otimes y_j = 0 \text{ in } M \otimes_R N$$

is equivalent to the following: There exist $a_{i,j} \in R$ with $a_{i,j} = 0$ for all but finitely many pairs (i, j) such that

$$m_j = \sum\nolimits_{i\in I} a_{i,j} x_i \quad \text{for all } j \in J,$$
$$0 = \sum\nolimits_{j\in J} a_{i,j} y_j \quad \text{for all } i \in I.$$

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The sufficiency is immediate. Suppose that $\sum_{j\in J} m_j \otimes y_j = 0$. Consider the short exact sequence

$$0 \to K \to \bigoplus\nolimits_{j\in J} R \to N \to 0$$

where the jth basis vector of $\bigoplus_{j\in J} R$ maps to y_j. Tensor this with M to get the exact sequence

$$K \otimes_R M \to \bigoplus\nolimits_{j\in J} M \to N \otimes_R M \to 0.$$

The assumption implies that there exist elements $k_i \in K$ such that $\sum k_i \otimes x_i$ maps to the element $(m_j)_{j\in J}$ of the middle. Writing $k_i = (a_{i,j})_{j\in J}$ and we obtain what we want. $\square$